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SMOTE-Enhanced Machine Learning Models for Reliable Crop Recommendation System

Ahmet Aytuğ Ayrancı
Dept. of Electrical and Electronics Engineering
İstanbul Kültür University
İstanbul, Turkey
ahmet.ayranci@iku.edu.tr

Mert Hayar
Dept. of Electrical and Electronics Engineering
İstanbul Kültür University
İstanbul, Turkey
m.hayar@iku.edu.tr

Atilla Özmen
Dept. of Electrical and Electronics Engineering
İstanbul Kültür University
İstanbul, Turkey
a.ozmen@iku.edu.tr

***Abstract*— Agricultural productivity is mainly determined by the crop's compatibility with soil and climate conditions. Consequently, crop recommendations play a crucial role in smart agricultural applications. This study introduces a machine learning-based crop recommendation system developed to support smart agriculture with IoT-enabled sensor data, including soil nutrients (N, P, K), pH, temperature, and rainfall. Several machine learning algorithms, namely Multi-Layer Perceptron (MLP), k-Nearest Neighbor (k-NN), Decision Tree (DT), and Random Forest (RF), were implemented and evaluated using accuracy, precision, recall, f-score, Root Mean Square Error (RMSE), and Mean Absolute Error (MAE). The dataset initially contained 2,534 samples with an imbalanced class distribution; to address this issue, the Synthetic Minority Over-sampling Technique (SMOTE) was applied, expanding the dataset to 8,250 samples and providing a more balanced representation. Experimental results showed that tree-based models, particularly DT and RF, achieved superior predictive performance, exceeding %99 in accuracy and f-score while maintaining minimal error rates. The k-NN algorithm also demonstrated robust results, whereas MLP provided average but consistent outcomes. Furthermore, the application of SMOTE improved the overall stability of classification results, as confirmed by k-fold cross-validation, and underscored the role of preprocessing in improving model generalization. These findings indicate that integrating Machine Learning (ML) techniques with appropriate data balancing strategies can yield reliable and interpretable crop recommendation systems, contributing to optimized resource management, increased yields, and sustainable agricultural practices.**

***Keywords*— SMOTE, Crop Recommendation System, Smart Agriculture, Data Preprocessing, Machine Learning**

I. INTRODUCTION

The agricultural sector remains one of the largest components of the global economy, employing billions of individuals whose livelihoods are directly tied to it. Furthermore, the sector plays a critical role in meeting the growing global demand for food. Historically, the advancement of civilizations has been closely linked to progress in agricultural practices and technologies. The mechanization of agriculture, which began with the industrial revolution, has progressed rapidly with the integration of developing technologies into agriculture today. The integration of advanced technologies has resulted in a decrease in the labor force needed in agriculture, while also offering greater opportunities for comprehensive knowledge from the field.

To ensure high product quality and efficiency in agriculture, detailed information about the agricultural field and climate conditions is essential [1-2]. Various technologies, including sensors, drones, agricultural robots, and cameras, are extensively utilized to collect precise and detailed information on farming fields [3-4]. These technologies enable the monitoring of various environmental and crop conditions, facilitating more informed decision-making and improved management practices within the sector. The most widely used remote sensing technology in modern agriculture is IoT-based sensors. They are significantly useful because they are both cost-effective and it is possible to obtain detailed information about the condition of the soil and the plant with various sensors.

Crop recommendation systems play a pivotal role in modern agriculture by supporting informed decision-making processes. These systems assist farmers in determining the most suitable crops to cultivate based on a comprehensive analysis of environmental, agronomic, and climatic factors. Crop recommendation systems, supported by advanced technologies and data analytics [5–7], help optimize resource use, enhance yield quality, and limit unnecessary inputs, thereby contributing to both economic and environmental sustainability. The integration of autonomous machinery, AI-driven decision support systems, and precision farming techniques further enhances the efficiency of agricultural operations and minimizes waste throughout the production cycle.

Crop recommendation systems integrate diverse parameters such as soil properties, and climate conditions to provide accurate and site-specific guidance for farmers. For instance, in the study by [8], the authors developed a crop recommendation system using various Machine Learning (ML) algorithms on sensor-derived datasets collected in India. Their goal was to determine which local crops would provide the highest yield potential by analyzing soil and atmospheric characteristics. Similarly, in [9], the researchers investigated the relationship between soil composition and fertilizer types, aiming to identify the most suitable combinations for maximizing harvest quantity. These examples highlight the importance of integrating sensor technologies and intelligent decision-making mechanisms to accurately evaluate agronomic variables and support optimized crop planning.

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Modern agriculture faces increasing challenges due to climate change, including extreme weather events, rising temperatures, drought, and declining soil fertility [10]. These factors have heightened the need for innovative and adaptable agricultural models to ensure the sustainability of food production. AI-powered systems can assist farmers in optimizing planting schedules, irrigation management, and harvesting times, thereby enhancing overall productivity [11-12]. Moreover, precision agriculture techniques enable more efficient use of critical resources such as water and fertilizers, leading to reduced operational costs and minimized environmental impact. Such technology-driven adaptation strategies hold significant potential for securing the future of agricultural production in the face of environmental uncertainty.

In artificial intelligence-based applications, several parameters influence overall system performance. Data preprocessing techniques are among the most important of these parameters. The quality, integrity, and appropriate transformation of the dataset directly impact performance criteria such as accuracy, generalization, and learning speed of ML algorithms. Appropriately removing missing or inconsistent data, eliminating unnecessary features, addressing unbalanced data distribution, and correctly applying data scaling are critical to improving model effectiveness. Furthermore, steps such as converting categorical data to numerical form, correcting unbalanced data distribution, and feature engineering are key factors supporting the learning process. Therefore, for a successful AI application, data preprocessing must be planned and implemented as carefully as model selection.

In this study, a crop recommendation system designed to enhance productivity and achieve higher yields in modern agricultural practices is analyzed using various ML techniques. Section II provides details about the dataset employed in the study, while Section III presents background information on crop recommendation systems and the ML algorithms used. Section IV discusses the evaluation metrics and simulation results that reflect the performance of the algorithms. Finally, Section V concludes the study by summarizing the findings and highlighting key outcomes.

II. Crop Recommendation systems

Crop selection is a fundamental decision in agricultural planning, as it directly influences productivity, resource utilization, and economic return. However, traditional methods of crop selection often rely on farmers' experience or static regional guidelines, which may not fully account for dynamic environmental conditions, soil health, and evolving climatic patterns. To address this challenge, crop recommendation systems have emerged as intelligent tools designed to assist farmers in choosing the most suitable crops for their specific agro-climatic and soil conditions.

The proposed crop recommendation system leverages real-time data collected from IoT-enabled sensors deployed in the field, such as soil pH, ambient temperature, and other relevant parameters [13-14]. This data-driven approach enables a more precise and adaptive decision-making process. The core objective of the system is to analyze local field conditions and generate crop suggestions that optimize yield potential, minimize resource wastage, and support sustainable farming practices. Crop recommendation systems play a crucial role in smart agriculture applications, particularly in addressing the challenges posed by a growing global population, extreme weather events driven by climate change, and yield losses caused by pests and diseases. By providing data-driven guidance on optimal crop selection, these systems help farmers make informed decisions that enhance productivity and resource efficiency. Moreover, such systems contribute to risk mitigation and resilience by adapting agricultural practices to dynamic environmental and biological stressors.

The integration of a crop recommendation system into smart farming practices represents a significant step toward precision agriculture, where technology assists in making context-aware, efficient, and sustainable farming decisions. While this study focuses primarily on the system's design and data-driven capabilities, detailed discussion of the ML methodology and evaluation metrics are provided in subsequent sections.

III. Dataset Information

Establishing a reliable and accurate dataset is fundamental for the effectiveness of AI-based decision-making systems. High-quality data not only enhances the performance of these systems but also ensures that the decisions they generate are reliable. In this study, two datasets from the Kaggle open-source data repository were combined using data mining techniques. The new data set obtained contains a total of 2,534 data entries, which have a class imbalance that can affect the performance of classification algorithms. To address the adverse effects of this imbalance and enhance model performance, the Synthetic Minority Over-sampling Technique (SMOTE) was implemented [15]. By using this technique, the sample size increased from 2,534 to 8,250.There are seven distinguishing features in the data set used in the study. The first three of these features are Nitrogen (N), Phosphorus (P), and Potassium (K), which indicate the nutrient values in the soil. The pH value is a critical parameter in assessing soil quality. A pH value of less than 7 signifies acidic soil conditions, which are conducive to the growth of certain crops that thrive in such environments. On the other hand, a pH value exceeding 7 indicates basic (alkaline) soil, which is suitable for cultivating plant species that prefer alkaline conditions. The output classes of the dataset consist of different crop types, including cotton, maize, rice, grapes, apple, banana, coconut, coffee, mango, orange, and papaya. These classes represent the crops that achieve the highest yield under given soil and climatic conditions. Table I shows the number of samples corresponding to each crop class in the original dataset, highlighting the uneven distribution of data across categories.

As shown in Table I, the dataset exhibits an imbalanced class distribution. Data imbalance is a major issue that affects ML model performance; therefore, to solve this problem, new data samples were generated. In this study, SMOTE was used to reduce potential biases and instability resulting from this imbalance. Through this technique, synthetic data was generated, and the dataset was expanded to 750 samples for each class, resulting in a total of 8,250 instances. This balanced structure helps the classifiers learn more representative patterns from each class. As a result, the overall reliability and robustness of the prediction models are significantly improved.

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TABLE I. DATASET DISTRIBUTION

Crop Type	Number of Crop
Cotton	750
Maize	450
Rice	409
Grapes	225
Apple	100
Banana	100
Coconut	100
Coffee	100
Mango	100
Orange	100
Papaya	100

IV. METHODOLOGY

In this section, detailed information is provided regarding the ML algorithms and oversampling technique employed in the study, with an emphasis on their function within the crop recommendation system.

A. Data Preprocessing

Data preprocessing techniques are crucial in ML-based applications. These techniques help ML models to recognize meaningful patterns in the dataset and produce more reliable predictions [16]. In agricultural datasets, sensor readings can differ widely in scale, and distribution, which makes tailored preprocessing methods necessary. Imbalanced class distributions, in particular, can lead to biased models that favor majority classes while underrepresenting minority ones. To address this issue, an oversampling technique was employed in this study to mitigate the effects of class imbalance, thereby enhancing the model's ability to make accurate and fair predictions across all crop types. Furthermore, the performance of the dataset expanded through oversampling was compared with the performance of the original dataset and the ML models, and the performance of this process performed in the data preprocessing step was analyzed.

The dataset used in this study comprises 2,534 distinct data **points**, each representing a unique set of measurements obtained through IoT-based sensors. The data includes **six** discriminative features—serving as input variables—and one target feature representing the crop classification label. These features form the foundation for training and evaluating the ML models implemented in the proposed crop recommendation system. In the study, the performance of the original dataset was tested with various ML algorithms and then data was generated using the oversampling technique to prevent unbalanced data distribution within the dataset. Specifically, the SMOTE was applied to produce synthetic data for the minority classes. The performance of this newly balanced dataset was then analyzed using the same ML algorithms. Finally, a comparative analysis was conducted between the results obtained from the original dataset and those derived from the SMOTE-balanced dataset, with the aim of evaluating the impact of oversampling on model performance.

B. Multi-Layer Perceptron

Multi-Layer Perceptron (MLP) is a class of feedforward Artificial Neural Network (ANN) that consists of an input layer, one or more hidden layers, and an output layer. Each layer is composed of interconnected nodes (neurons), where each node in one layer is fully connected to the nodes in the subsequent layer. MLPs are capable of learning complex, non-linear relationships between input features and output targets through a supervised learning process. Each neuron in an MLP computes a weighted sum of its input signals and passes the result through a non-linear activation function. Commonly used activation functions include the sigmoid, hyperbolic tangent (tanh), and the Rectified Linear Unit (ReLU) [17]. In MLP, activation functions play a crucial role in introducing non-linearity into the network, thereby enabling it to learn complex patterns and approximate arbitrary functions. Without activation functions, an MLP, regardless of its depth, would behave like a simple linear model due to the composition of linear transformations remaining linear. Mathematically, for a given input vector x, weight vector w, and bias b, the neuron computes:

$$z = w^T x + b \tag{1}$$

$$a = \emptyset(z) \tag{2}$$

This process is repeated across all neurons and layers within the MLP, allowing the model to learn complex, non-linear mappings from sensor inputs to recommended crop outputs. Activation functions introduce non-linearity into the model, enabling the MLP to approximate highly complex functions. The training of MLPs is typically conducted using the backpropagation algorithm, which computes the gradient of the loss function with respect to each weight in the network. The weights can be updated iteratively using optimization techniques such as Stochastic Gradient Descent (SGD), RMSprop, and Adam.

In recent years, MLPs have been widely adopted in smart systems due to their ability to model complex, non-linear patterns from sensor data. In agricultural applications, they are particularly effective for tasks such as crop recommendation, yield prediction, and disease detection, where multiple environmental variables must be analyzed simultaneously. Their adaptability and learning capability make MLPs a powerful tool in the development of intelligent, IoT-enabled precision agriculture systems.

C. K-Nearest Neighbor

The k-Nearest Neighbors (k-NN) algorithm is a non-parametric, instance-based learning method that is widely used for classification and regression tasks. Originally introduced by Fix and Hodges in 1951 and later formalized by Cover and Hart in 1967, k-NN operates by identifying the k closest training samples (neighbor) to a given input and assigning a prediction based on classification [18].

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In this study, the k-NN algorithm is utilized to recommend suitable crops by analyzing multidimensional environmental data collected from IoT-enabled sensors, including minerals from soil, pH, temperature and rainfall. Euclidean distance is used as the primary distance metric to quantify the similarity between data points. The choice of the *k* parameter and the distance metric significantly influences the performance of the k-NN algorithm. Among various distance measures, Euclidean, Manhattan, and Minkowski distances are the most commonly utilized in the literature due to their computational efficiency and adaptability to different data distributions. Selecting an appropriate distance function is crucial, especially in high-dimensional or heterogeneous sensor datasets, as it directly affects the model's ability to distinguish between similar and dissimilar instances. The suitability of k-NN for this application stems from its simplicity, ability to capture non-linear relationships, and robustness against noisy and heterogeneous sensor data.

D. Decision Tree

The Decision Tree (DT) algorithm is a supervised learning method widely used for both classification and regression tasks due to its interpretability and low computational complexity. It operates by recursively partitioning the dataset into subsets based on features, forming a tree-like structure where each internal node represents a decision rule on a particular feature, and each leaf node corresponds to an output class. The algorithm selects features and splitting thresholds by optimizing criteria such as Gini impurity or information gain, with the goal of maximizing class homogeneity within the branches. Its hierarchical structure allows the model to learn simple decision rules that are easy to visualize and interpret, making it particularly useful in decision-making applications like agricultural planning.

The DT algorithm is well-suited for the dataset used in the study, particularly for classifying agricultural data that includes soil nutrient concentrations (N, P, K) alongside environmental features such as pH, temperature, and rainfall. These features often exhibit non-linear and hierarchical relationships in determining crop suitability. DT can effectively capture such interactions through a series of recursive, rule-based splits that require no assumptions about data distribution. In this study, DT is employed to identify interpretable decision paths that map specific combinations of soil and environmental parameters to suitable crop types. This interpretability is particularly valuable in agricultural contexts, as it enables farmers and agronomists to understand and trust the basis of the model's recommendations.

E. Random Forest

Random Forest (RF) is an ensemble-based ML algorithm introduced by Leo Breiman in 2001 as an improvement over traditional DT models [19]. It combines the predictions of multiple DTs, each trained on a randomly sampled subset of the data and features, to produce a more accurate and robust overall model. This approach significantly reduces the risk of overfitting and improves generalization performance by leveraging the power of "bagging" and feature randomness. Since its introduction, RF has become a widely used algorithm in various domains, particularly where interpretability, robustness, and performance are all critical. The main advantage of RF is that it provides higher accuracy than a single DT and significantly reduces the tendency for overfitting.

In smart agriculture applications, the RF algorithm is preferred for many important problems due to its ability to work with complex and noisy data. It demonstrates superior performance in processing multidimensional data, particularly in sensor-based agricultural systems [20]. In this context, it yields effective results in various tasks, including soil property estimation, irrigation scheduling, fertilization recommendations, crop yield estimation, and early detection of plant diseases. The clarity offered by the RF model makes it easier for agricultural experts and farmers to see how and why certain predictions are made. By showing which factors influence its decisions, it turns complex data into practical insights they can trust and act on. With all these aspects, RF is considered a reliable, effective, and scalable tool for the development of artificial intelligence-supported agricultural decision support systems.

F. K-Fold Cross Validation

K-fold cross-validation is a robust validation technique commonly used to independently and reliably evaluate the performance of ML models [21]. It is particularly effective for assessing the performance of classification models in scenarios where the dataset contains a large number of target classes but a limited number of data samples. In standard ML practice, the dataset is typically partitioned into two subsets, training and testing, according to a predetermined ratio. Some studies extend this approach by incorporating a validation set, thereby dividing the data into three subsets. When the dataset is split into training and testing only once, the resulting performance metrics are inherently dependent on that particular partition, which may lead to biased or unstable estimates. To address this limitation, the k-fold cross-validation technique partitions the dataset into *k* equally sized folds in a randomized manner. The model is then trained and evaluated *k* times, each time using a different fold as the test set and the remaining k-1 folds as the training set. The overall performance is obtained by aggregating the evaluation metrics across all folds, providing a more reliable and robust estimate of the model's generalization ability.

V. PERFORMANCE METRICS AND EVOLUATION

To assess the effectiveness of the ML algorithms employed in this study, several widely accepted performance metrics have been used, including accuracy, precision, recall, and f-score. These metrics provide a comprehensive understanding of how well each model performs in classifying crop types based on sensor-derived features.

A. Performance Metrics

In this study, different performance metrics are used to assess the effectiveness and reliability of the ML models in AI-driven applications. These metrics offer a quantitative basis for evaluating predictive performance. In crop recommendation systems that rely on IoT-enabled sensor data, they are particularly important for verifying the model's capacity to generalize under diverse agricultural conditions.

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Accuracy, indicates the proportion of correct classifications achieved by the ML algorithm on the dataset.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

TP refers to True Positive, TN refers to True Negative, FP refers to False Positive, and FN refers to False Negative parameters.

Precision, indicates the success rate in correctly identifying instances predicted as positive and represents the proportion of correctly predicted positive cases to the total number of cases predicted as positive.

$$Precision = \frac{TP}{TP + FP} \quad (4)$$

Recall, represents the proportion of actual positive instances that are correctly identified as positive by the model, relative to the total number of true positive cases.

$$Recall = \frac{TP}{TP + FN} \quad (5)$$

F-Score, allows the simultaneous measurement of precision and recall. It provides the harmonic mean of these values.

$$F - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (6)$$

Root Mean Square Error (RMSE) is a quadratic metric that quantifies the average magnitude of the difference between predicted and actual values in a dataset.

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2} \quad (7)$$

Where, n denotes the total number of observations, $\hat{y}_i$ represents the predicted value for the i-th observation, and y_i represents the actual value for the i-th observation, with the formula measuring the magnitude of the error by taking the square root of the mean of the squared differences between predicted and actual values.

Mean Absolute Error (MAE) is the average horizontal distance between two continuous variables. It represents the mean distance between all actual values and the value that best fits the data. As an easily interpretable metric, MAE is widely used in both regression and classification problems. It is negatively oriented, meaning that estimators with lower MAE values demonstrate higher performance. The MAE value can be calculated as follows.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \quad (8)$$

RMSE and MAE are error-based metrics, where smaller values reflect higher predictive accuracy of the model.

VI. Simulation Results

In this section, the performances of the ML models used in the study are examined. The primary objective is to compare the success of the models using various performance metrics such as accuracy, precision, and recall. Furthermore, simulation experiments were conducted on both the original imbalanced dataset and the SMOTE-balanced dataset to evaluate the impact of data balancing on classification performance. This dual analysis allows for a comprehensive assessment of how oversampling influences model robustness and generalization capability.

TABLE II. ML Performance without oversampling

Algorithms	Accuracy	Precision	Recall	F-Score
MLP	%87.05	%87.2	%87	%86.7
k-NN	%95.9	%96	%95.9	%95.9
DT	%97	%97.1	%97.1	%97
RF	%97.65	%97.7	%97.7	%97.6

Table II shows that without oversampling, the performance of most algorithms declines, particularly for those sensitive to class imbalance. The k-NN, DT, and RF still maintain relatively high performance, with accuracy, precision, recall, and f-score values above 95%, indicating a degree of robustness to imbalance. MLP experiences moderate performance, reflecting a pronounced difficulty in correctly classifying minority classes.

TABLE III. ML Performance with oversampling

Algorithms	Accuracy	Precision	Recall	F-Score
MLP	%89.81	%90	%89.8	%89.5
k-NN	%99.27	%99.3	%99.3	%99.3
DT	%99.13	%99.1	%99.1	%99.1
RF	%99.52	%99.5	%99.5	%99.5

Table III demonstrates that, with oversampling applied, all algorithms except MLP achieve exceptionally high performance, with k-NN, DT, and RF exceeding %99 in accuracy, precision, recall, and f-score, indicating highly consistent and balanced classification capabilities. MLP also performs well, maintaining close to %90 across all metrics, still benefits from the oversampling process.

A comparison of Table II and Table III demonstrate a consistent improvement in classification performance across all algorithms following the application of oversampling. Performance evaluations with the SMOTE-balanced dataset showed that all ML models achieved improved classification performance. Thus, these results confirm that oversampling is an effective approach to addressing unbalanced class distributions, leading to significant improvements in predictive performance, particularly in algorithms that are more vulnerable to imbalance.

TABLE IV. K-Fold effects on ML algorithm without oversampling

Algorithms	K=2	K=3	K=5	K=10
MLP	%81.61	%84.14	%84.85	%83.26
k-NN	%92.74	%95	%96.17	%96.37
DT	%96.17	%96.20	%98	%97.87
RF	%97.4	%97.47	%98.58	%98.62

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Table IV shows that without oversampling, k-fold cross-validation accuracy values are generally lower, especially for algorithms that are more sensitive to class imbalance. The k-NN, DT, and RF still show competitive performance, maintaining accuracy above %94 for all k-fold analysis, with slight improvements as k value increases. The MLP achieved moderate performance (%81.61–%83.26), indicating considerable difficulty in addressing the effects of imbalanced data.

TABLE V. K-FOLD EFFECTS WITH OVERSAMPLING

Algorithms	K=2	K=3	K=5	K=10
MLP	%91.75	%90.76	%89.98	%91.05
k-NN	%98.50	%99.05	%99.15	%99.20
DT	%98.80	%99.05	%99.15	%99.26
RF	%99.10	%99.44	%99.54	%99.52

Table V indicates that with oversampling applied, all algorithms maintain consistently high accuracy across different k-fold values. The k-fold cross-validation results on the SMOTE-balanced dataset show a substantial performance improvement across all ML algorithms used. These outcomes indicate that oversampling not only addresses class imbalance but also enhances the consistency and reliability of model performance in crop recommendation applications.

A comparative analysis of these two tables indicates that the application of oversampling improves model performance across all algorithms and K-fold values. MLP exhibits the most evident improvement, with performance increasing from approximately %81-%83 to above %91 at higher fold values. Algorithms such as k-NN, DT, and RF also show improved accuracy with oversampling, reaching almost perfect accuracy compared to their slightly lower, although still high, results in the absence of oversampling. Overall, the results suggest that oversampling not only enhances classification accuracy but also reduces variability across k-fold evaluations, thereby improving model stability and generalization particularly for algorithms more sensitive to class imbalance.

TABLE VI. ERROR RATES OF ML ALGORITHMS

Algorithms	RMSE	MAE
MLP	%14	%3.25
k-NN	%8.4	%0.82
DT	%6.66	%0.7
RF	%6	%1

Table VI shows error rates of the ML algorithms. RF and DT have the lowest error values, reflecting their high accuracy and robustness. The k-NN algorithm also performed well, with comparatively low error rates. By contrast, the MLP yielded higher error values, indicating greater variability in the classification process. These findings indicate that tree-based methods provide more reliable and accurate results for this dataset, especially in achieving lower error levels.

VII. CONCULUSION AND FUTURE WORKS

This study presents a ML-based crop recommendation system aimed at supporting smart agriculture applications. By utilizing soil nutrients and environmental parameters, ML models were applied to determine the most suitable crop. The results indicated that tree-based methods, specifically DT and RF, consistently achieved significantly high performance levels, as shown by their superior accuracy, precision, recall, f-score, and the lowest error rates (RMSE and MAE). The k-NN model also exhibited superior predictive accuracy, while MLP yielded comparatively moderate results.

The implementation of the SMOTE was shown to significantly improve the performance of all models by mitigating class imbalance and enhancing the consistency of results across multiple k-fold validations. This emphasizes the importance of data preprocessing in improving model robustness and ensuring fair representation of all classes.

Overall, the findings confirm that integrating ML algorithms with effective data preprocessing strategies can provide reliable and accurate decision support tools for crop recommendations. Such systems have the potential to guide farmers in improving yields and promoting sustainable agricultural applications. Future work aims to contribute to the literature by using deep learning models and explainable AI methods on crop recommendation systems.

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